

PHYSICS

SECTION-1

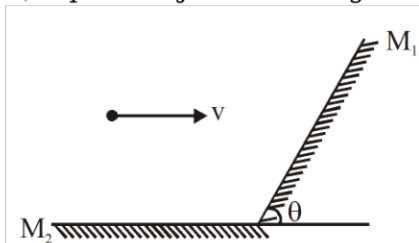
1) For Fraunhofer single slit diffraction -

- (1) width of central maxima is proportional to λ
- (2) on increasing the slit width, the width of central maxima decreases
- (3) on making the slit width $a = \lambda$, central fringe spreads in the range $\pm 90^\circ$
- (4) all of the above are correct

2) Select the correct statement.

- (1) Concave mirror always form real image for real object
- (2) Concave mirror always form virtual image for real object
- (3) Concave mirror always form real image for virtual object
- (4) Convex mirror always form virtual image for virtual object

3) A point object is moving with a speed v before an arrangement of two mirrors as shown in figure.



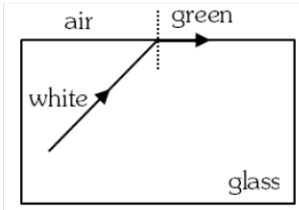
Find the velocity of image in mirror M_1 respect to image in mirror M_2

:-

- (1) $2V \sin \theta$
- (2) $V \sin \theta$
- (3) $2V \cos \theta$
- (4) $V \cos \theta$

4) In a compound microscope, if the objective produces an image I_o and eye piece produces an image I_e , then -

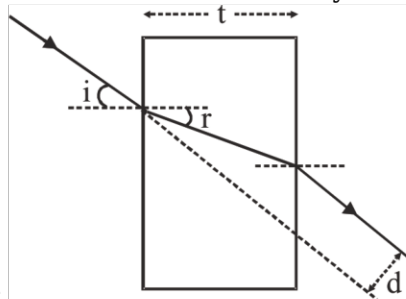
- (1) I_o virtual but I_e is real
- (2) I_o is real but I_e is virtual
- (3) I_o and I_e are both real
- (4) I_o and I_e are both virtual



5) Which set of colours will come out in air for a situation shown in figure?

- (1) Yellow, Orange and Red
- (2) All
- (3) Orange, Red and Violet
- (4) Blue, Green and Yellow

6) A ray of light is incident on a thick slab of glass of thickness t as shown in the figure. The emergent ray is parallel to the incident ray but displaced sideways by a distance d . If the angles are



small then d is. :-

- (1) $t \left(1 - \frac{i}{r} \right)$
- (2) $rt \left(1 - \frac{i}{r} \right)$
- (3) $it \left(1 - \frac{r}{i} \right)$
- (4) $t \left(1 - \frac{r}{i} \right)$

7) Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this :

- (1) $\sin^{-1} \left(\frac{10t_1}{t_2} \right)$
- (2) $\sin^{-1} \left(\frac{t_2}{t_1} \right)$
- (3) $\sin^{-1} \left(\frac{10t_2}{t_1} \right)$
- (4) $\sin^{-1} \left(\frac{t_1}{10t_2} \right)$

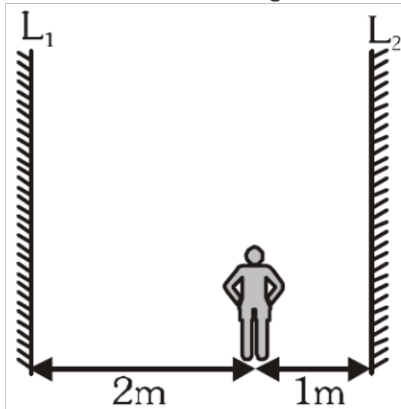
8) A microscope is focused on a mark, then a glass slab of refractive index 1.5 and thickness of 6 cm is placed on the mark to get the mark again in focus, the microscope should be moved

- (1) 4 cm
- (2) 2 cm

(3) 6 cm

(4) 8 cm

9) Two mirrors labelled L_1 for left mirror and L_2 for right mirror in the figure are parallel to each other and 3.0 m apart. A person standing 1.0 m from the right mirror (L_2) looks into this mirror and sees a series of images. The second nearest image seen in the right mirror is situated at a distance :



(1) 2.0 m from the person

(2) 4.0 m from the person

(3) 6.0 m from the person

(4) 8.0 m from the person

10) A fish looking up through water sees the outside world contained in a circular horizon. If

refractive index of water is $\frac{4}{3}$ and fish is 12 cm below the surface, radius of circle is :-

(1) $12 \times 3 \times \sqrt{5}$ cm(2) $12 \times 3 \times \sqrt{7}$ cm(3) $12 \times \sqrt{\frac{5}{3}}$ cm(4) $\frac{12 \times 3}{\sqrt{7}}$ cm

11)

The angle of minimum deviation δ_m for an equilateral glass prism is 30° . Refractive index of the prism is:-

(1) $1/\sqrt{2}$ (2) $\sqrt{2}$ (3) $2\sqrt{2}$ (4) $1/2\sqrt{2}$

12) The objective lens of a compound microscope produces magnification of 10. In order to get an overall magnification of 100 when image is formed at 25 cm from the eye, the focal length of the eye piece lens should be :-

- (1) 4 cm
- (2) 10 cm
- (3) $\frac{25}{9}$ cm
- (4) 9 cm

13) A compound microscope having $f_o = 7$ mm, $f_e = 20$ mm and distance between the lenses is 20 cm. If image is formed at least distance of distinct vision then magnifying power?

- (1) 337
- (2) 400
- (3) 220
- (4) 120

14) The refracting angle of a prism is A , and refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is

- (1) $90^\circ - A$
- (2) $180^\circ + 2A$
- (3) $180^\circ - 3A$
- (4) $180^\circ - 2A$

15) The minimum distance between the real object and its real image for concave mirror is

- (1) f
- (2) $2f$
- (3) $4f$
- (4) Zero

16) In one face of prism is silvered having prism angle 30° & $\mu = \sqrt{2}$. What will be the angle of incidence so that the incidence ray retraces its path.

- (1) 30°
- (2) 60°
- (3) 90°
- (4) 45°

17) A clock hung on a wall has marks instead of numerals on its dial. On the adjoining wall, there is a plane mirror and the clock indicates the time 4 : 50. Then the time on the image of clock is

- (1) 7 : 10
- (2) 4 : 50
- (3) 5 : 40
- (4) 10 : 07

18) An air bubble in a glass slab with refractive index 1.5 (near normal incidence) is 5cm deep when

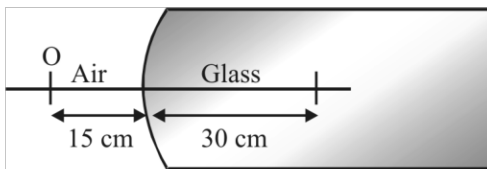
viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness of slab is

- (1) 8 cm
- (2) 10 cm
- (3) 12 cm
- (4) 16 cm

19) A ray is incident at an angle of incidence i on one surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , the angle of incidence i is nearly equal to :

- (1) A/μ
- (2) $A/2\mu$
- (3) μA
- (4) $\mu A/2$

20) A point object O is placed in front of a glass rod ($\mu = 1.5$) having spherical end of radius of curvature 30 cm. The image would be formed at :-



- (1) 30 cm left
- (2) Infinity
- (3) 1 cm to the right
- (4) 18 cm to the left

21) An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eye piece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and the focal length f_e of the eye piece are :-

- (1) $f_o = 45$ cm and $f_e = -9$ cm
- (2) $f_o = 7.2$ cm and $f_e = 5$ cm
- (3) $f_o = 50$ cm and $f_e = 10$ cm
- (4) $f_o = 30$ cm and $f_e = 6$ cm

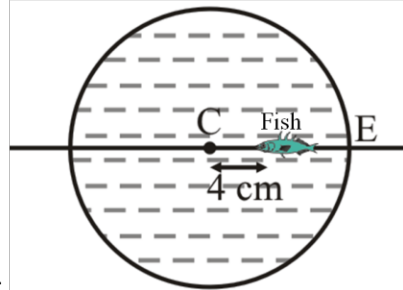
22) An infinitely long rod lies along the axis of a concave mirror of focal length f . The near end of the rod is at distance $u > f$ from the mirror. Its image will have a length-

- (1) $\frac{uf}{u-f}$
- (2) $\frac{uf}{u+f}$

(3) $\frac{f^2}{u+f}$

(4) $\frac{f^2}{u-f}$

23) In a thin spherical fish bowl of radius 10 cm filled with water of refractive index $\frac{4}{3}$ there is a small fish at a distance of 4 cm from the centre C as shown in figure. Where will the image of fish



appear, if seen from E :-

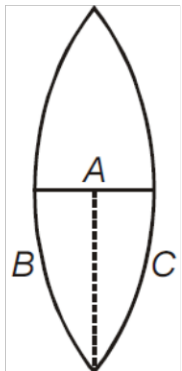
- (1) 5.2 cm
- (2) 7.2 cm
- (3) 4.2 cm
- (4) 3.2 cm

24) A point object is placed on the principal axis of a concave mirror quite far away from the pole and moved at a constant speed 0.5 cm/sec towards the pole. The image also moves. It is found that the object and the image cross each other at a point which is at a distance 50 cm from the pole. Position of the image when the object is at a distance 30 cm from the pole is:

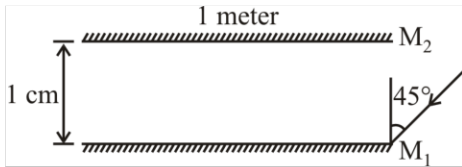
- (1) 150 cm from the pole in front of the reflecting surface
- (2) 120 cm from the pole behind the mirror
- (3) 150 cm from the pole behind the mirror
- (4) 120 cm from the pole in front of the reflecting surface

P_1, P_2

25) A thin equiconvex lens of power P is cut into three parts A, B and C as shown in figure. If P_1 and P_3 are powers of the three parts respectively, then



- (1) $P_1 = P_2 = P_3$
- (2) $P_1 > P_2 = P_3$
- (3) $P_1 < P_2 = P_3$
- (4) $P_2 = P_3 = 2P_1$



26) Two plane mirrors M_1 and M_2 each have length 1 m and are separated by 1 cm. A ray of light incident on one end of mirror M_1 at an angle of 45° . How many reflections will the ray have before going out from the other end ? (Including the first one)

- (1) 50
- (2) 51
- (3) 101
- (4) 104

27) The radius of curvature of a biconvex (equiconvex) lens is 40 cm and refractive index $\mu_g = 1.5$. Then the focal length of the lens is

- (1) 40cm
- (2) 50cm
- (3) -30cm
- (4) -40cm

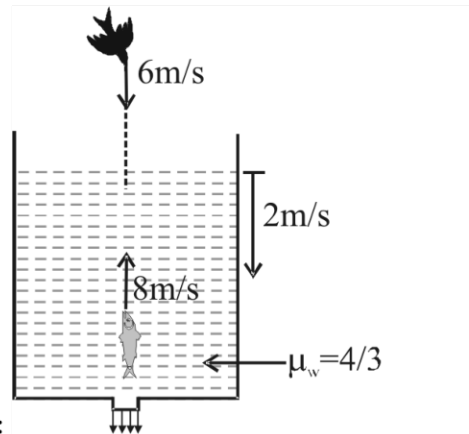
28) An object 25 cm high is placed in front of convex lens of focal length 30 cm. If the height of image formed is 50 cm, find the distance between real object and real image :-

- (1) 90 cm
- (2) 45 cm
- (3) 100 cm
- (4) 135 cm

29) A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a speed of 15 m/sec. The speed of the image of the second car as seen in the mirror by the first one will be :-

- (1) $\frac{1}{10}$ m/s
- (2) $\frac{1}{15}$ m/s
- (3) 10 m/s
- (4) 15 m/s

30) As shown in the figure, bird is moving downward with a speed of 6 m/s and fish is moving upward with a speed of 8 m/sec. At the instant shown, the water level falls at the rate of 2m/sec.



Velocity of fish, as seen by bird is:

- (1) $\frac{11}{2}\hat{j}$
- (2) $\frac{23}{2}\hat{j}$
- (3) $\frac{15}{2}\hat{j}$
- (4) Zero

31) The focal lengths of the objective and eye piece of a telescope are respectively 100 cm and 2 cm. The moon subtends an angle of 0.5° at the eye. If it is looked through the telescope, the angle subtended by the moon's image will be

- (1) 100°
- (2) 50°
- (3) 25°
- (4) 10°

32) A plano convex lens fits exactly into a plano concave lens. Their plane surfaces are parallel to each other. If lens are made of different materials of refractive indices μ_1 and μ_2 & R is the radius of curvature of curved surface of the lenses, then the focal length of combination is:-

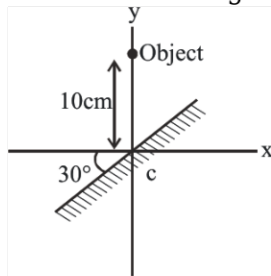
- (1) $\frac{2R}{\mu_2 - \mu_1}$
- (2) $\frac{R}{2(\mu_2 + \mu_1)}$
- (3) $\frac{R}{2(\mu_1 - \mu_2)}$
- (4) $\frac{R}{\mu_1 - \mu_2}$

33) Concave and convex lenses are placed touching each other. The ratio of magnitudes of their focal length is 2 : 3. The focal length of the system is +30 cm. Then the focal lengths of individual lenses are:-

- (1) - 75, 50

- (2) - 15, 10
 (3) 75, 50
 (4) 75, - 50

34) The co-ordinates of the image with respect to the origin c will be



- (1) $(-5, 5\sqrt{3})$
 (2) $(5, -5\sqrt{3})$
 (3) $(5\sqrt{3}, -5)$
 (4) None of these

35) A convex lens of focal length 12 cm is made of glass of $\mu = \frac{3}{2}$. What will be its focal length when immersed in liquid of $\mu = \frac{5}{4}$?

- (1) 6 cm
 (2) 12 cm
 (3) 24 cm
 (4) 30 cm

SECTION-2

1) A light beam is being reflected by using two mirrors, as in a periscope used in submarines. If one of the mirrors rotates by an angle θ , the reflected light will deviate from its original path by the angle

- (1) 2θ
 (2) 0°
 (3) θ
 (4) 4θ

2) In a concave mirror experiment, an object is placed at a distance x_1 from the focus and the image is formed at a distance x_2 from the focus. The focal length of the mirror would be :-

- (1) $x_1 x_2$
 (2) $\sqrt{x_1 x_2}$
 (3) $\frac{x_1 + x_2}{2}$

(4) $\sqrt{\frac{x_1}{x_2}}$

3) A light wave is incident normally over a slit of width 24×10^{-5} cm. The angular position of second dark fringe from the central maxima is 30° . What is the wave length of light -

- (1) 6000 Å
- (2) 5000 Å
- (3) 3000 Å
- (4) 1500 Å

4) A polaroid is placed at 45° to an incoming unpolarised light of intensity I_0 . Now the intensity of light passing through the polaroid after polarisation would be -

- (1) I_0
- (2) $I_0/2$
- (3) $I_0/4$
- (4) zero

5) The height up to which water should be filled in a container of height 21 cm so that it appears half filled when seen from the above is (refractive index of water is $4/3$) :-

- (1) 10 cm
- (2) 12 cm
- (3) 13 cm
- (4) 14 cm

6) A ray of light is incident along a vector $\hat{i} + \hat{j} - \hat{k}$ on a plane mirror lying in x-y plane. The unit vector along the reflected ray may be :

- (1) $\frac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$
- (2) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- (3) $\frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- (4) $\frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$

7) Two plane mirrors are at 45° to each other. If an object is placed between them, then the number of image will be :-

- (1) 5
- (2) 9
- (3) 7

(4) 8

8) A thin mica sheet of thickness 2×10^{-6} m and refractive index ($\mu = 1.5$) is introduced in the path of the upper wave. The wavelength of the wave used is $5000 \text{ \AA}$. The central bright maximum will shift

- (1) 2 fringes upward
- (2) 2 fringes downward
- (3) 10 fringes upward
- (4) None of these

9) In Young's double slit experiment, the intensity on the screen at a point where path difference is λ is K. What will be the intensity at the point where path difference is $\lambda/4$

- (1) $\frac{K}{4}$
- (2) $\frac{K}{2}$
- (3) K
- (4) Zero

10) Match the corresponding entries of Column-I with Column-II (Where m is the magnification produced by the mirror)

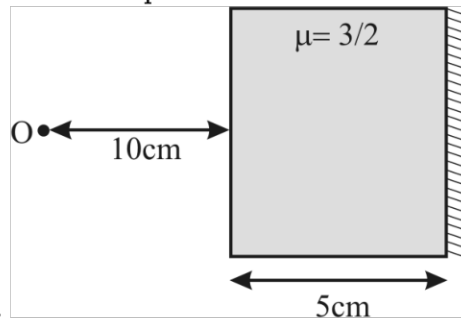
Column-I		Column-II	
(A)	$m = -2$	(a)	Convex mirror
(B)	$m = -\frac{1}{2}$	(b)	Concave mirror
(C)	$m = +2$	(c)	Real image
(D)	$m = +\frac{1}{2}$	(d)	Virtual image

- (1) $A \rightarrow b$ and c , $B \rightarrow b$ and c , $C \rightarrow b$ and d , $D \rightarrow a$ and d
- (2) $A \rightarrow a$ and c , $B \rightarrow a$ and d , $C \rightarrow a$ and b , $D \rightarrow c$ and d
- (3) $A \rightarrow a$ and d , $B \rightarrow b$ and c , $C \rightarrow b$ and d , $D \rightarrow b$ and c
- (4) $A \rightarrow c$ and d , $B \rightarrow b$ and d , $C \rightarrow b$ and c , $D \rightarrow a$ and d

11) In YDSE, two slits are separated by 3mm and illuminated by light 480 nm. The screen is at a distance of 2m from the plane of slit. Calculate the separation between 3rd bright fringe and the 1st dark fringe:-

- (1) 0.8 mm
- (2) 0.4 mm
- (3) 0.6 mm
- (4) 0.5 mm

12) A point object is placed in front of a thick plane mirror as shows in figure below. Find the

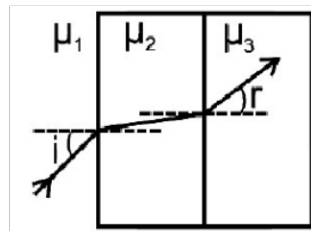


distance of final image from object.

- (1) $15/2$ cm
- (2) 15 cm
- (3) $40/3$ cm
- (4) $80/3$ cm

13) The phase difference between incident wave and reflected wave is 180° when light ray :

- (1) incident on glass from air
- (2) incident on air from glass
- (3) incident on water from glass
- (4) none of these



14) In the figure shown $\frac{\sin i}{\sin r}$ is equal to:

- (1) $\frac{\mu_2^2}{\mu_3 \mu_1}$
- (2) $\frac{\mu_3}{\mu_1}$
- (3) $\frac{\mu_3 \mu_1}{\mu_2^2}$
- (4) None of these

15) A ray of light is incident upon an air/water interface (it passes from air into water) at an angle of 45° . Which of the following quantities change as the light enters the water?

- (I) wavelength
- (II) frequency
- (III) speed of propagation
- (IV) direction of propagation

- (1) I, III only
- (2) III, IV only
- (3) I, II, IV only
- (4) I, III, IV only

CHEMISTRY

SECTION-1

1) Which forms interstitial compounds?

- (1) Fe
- (2) Co
- (3) Ni
- (4) All

2) For a catalyst, which condition is not essential:-

- (1) Variable valency
- (2) High ionisation energy
- (3) Empty orbital
- (4) Free valency on the surface

3) Which of the following ion convert orange solution of $K_2Cr_2O_7$ into green solution ?

- (1) Na_2SO_3
- (2) Na_2S
- (3) NaI
- (4) All of these

4) Which of the following electronic configuration can be associated with the highest spin only magnetic moment ?

- (1) $[Ar] 3d^8$
- (2) $[Ar] 3d^3$
- (3) $[Ar] 3d^6$
- (4) $[Ar] 3d^7$

5) Lanthanoid contraction is caused due to :-

- (1) The same effective nuclear charge from Ce to Lu
- (2) The imperfect shielding on outer electrons by 4f electrons from the nuclear charge
- (3) The appreciable shielding on outer electrons by 4f electrons from the nuclear charge
- (4) The appreciable shielding on outer electrons by 5d electrons from the nuclear charge

6) Which of the following solid salt on heating with solid $K_2Cr_2O_7$ and conc. H_2SO_4 orange red vapours are evolved which turn aqueous NaOH solution yellow ?

- (1) NaBr
- (2) NaCl

(3) NaNO_3 (4) NaI

7) Which of the following pair doesn't belong to transition elements ?

(1) Cu and Ag

(2) Cr and Mn

(3) Sc and Ti

(4) Zn and Cd

8) Which of the following complex compounds can exhibit highest paramagnetic behaviour ?

(1) Cr^{+3} (2) Mn^{+2} (3) V^{+5} (4) Mn^{+7}

9)

Which compound of Ti has magnetic moment 1.73 BM ?

(1) TiCl_2 (2) TiCl_3 (3) TiCl_4 (4) TiO_2

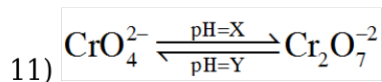
10) Elements of 3d series which does not show variable valency :-

(1) Fe, Zn

(2) Sc, Zn

(3) Mn, Zn

(4) Ti, Co



Values of (X) and (Y) are respectively

(1) 4 and 5

(2) 4 and 8

(3) 8 and 4

(4) 8 and 9

12) The aqueous solution containing which one of the following ions will be colourless ?

(1) Ti^{3+} (2) Mn^{2+}

(3) Sc^{3+}

(4) Fe^{2+}

13) Which of the following types of metals make the most efficient catalyst ?

(1) Transition metals

(2) Alkali metals

(3) Alkaline earth metals

(4) Coloured metals

14) Not more than one type oxidation state is shown by :-

(1) Mn

(2) Cr

(3) Fe

(4) Sc

15) Which of the following is a lanthanide :-

(1) Ta

(2) Rh

(3) Th

(4) Gd

16) The electronic configuration of ${}_{64}\text{Gd}$ is :-

(1) $[\text{Xe}] 6s^2 5d^0 4f^8$

(2) $[\text{Xe}] 6s^1 5d^1 4f^8$

(3) $[\text{Xe}] 6s^2 5d^1 4f^7$

(4) None of above

17) The oxidation state of 'Pt' in the complex $[\text{Pt}(\text{Py})_4][\text{PtCl}_4]$ are respectively :-

(1) 0 & 0

(2) +4 & +2

(3) +2 & +2

(4) 0 & +4

18) Which among the following is diamagnetic ?

(1) $[\text{Cr}(\text{NH}_3)_6]^{+3}$

(2) $[\text{CoF}_6]^{-3}$

(3) $[\text{Ni}(\text{CO})_4]$

(4) $[\text{CuCl}_4]^{-2}$

19) Among the following ions which one has the highest paramagnetic behaviour :

- (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{+3}$
- (2) $[\text{Cu}(\text{H}_2\text{O})_6]^{+2}$
- (3) $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$
- (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{+2}$

20) Primary and secondary valencies of Cobalt in $[\text{Co}(\text{EDTA})]^-$ are respectively

- (1) 2, 6
- (2) 3, 5
- (3) 2, 5
- (4) 3, 6

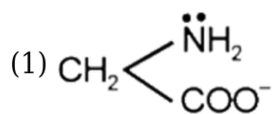
21) $[\text{Fe}(\text{NO}_2)_3\text{Cl}_3]$ and $[\text{Fe}(\text{ONO})_3\text{Cl}_3]$ are -

- (1) Linkage isomers
- (2) geometrical isomers
- (3) optical isomers
- (4) None of these

22) The IUPAC name for the complex compound $\text{Li}[\text{AlH}_4]$ is:

- (1) lithium aluminium hydride
- (2) hydrido-aluminiumlithium (III)
- (3) lithium tetrahydridoaluminate (III)
- (4) lithium tetrahydroxoaluminate (III)

23) Glycinato ligand is -



- (2) Bidentate ligand
- (3) Two donor sites N and O⁻
- (4) All of these

24) Give the correct increasing order of electrical conductivity of aqueous solutions of following complex entities

I	$[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$
II	$[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
III	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
IV	$\text{K}_2[\text{PtCl}_6]$

- (1) $\text{III} < \text{IV} < \text{II} < \text{I}$
- (2) $\text{IV} < \text{II} < \text{III} < \text{I}$
- (3) $\text{II} < \text{I} < \text{IV} < \text{III}$
- (4) $\text{I} < \text{II} < \text{IV} < \text{III}$

25) IUPAC name of $[\text{Pt}(\text{NH}_3)_3(\text{Br})(\text{NO}_2)\text{Cl}]\text{Cl}$ is -

- (1) Triamminechlorobromonitroplatinum(IV) chloride
- (2) Triamminebromonitrochloroplatinum(IV) chloride
- (3) Triamminebromidochloridonitroplatinum(IV) chloride
- (4) Triamminenitrochlorobromoplatinum(IV) chloride

26) Amongst $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$

- (1) $\text{Ni}(\text{CO})_4$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic
- (2) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CO})_4]$ paramagnetic
- (3) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ paramagnetic
- (4) $\text{Ni}(\text{CO})_4$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ paramagnetic

27) What is the shape of $\text{Fe}(\text{CO})_5$ molecule? Given that its dipole moment = 0.

- (1) Tetrahedral
- (2) Octahedral
- (3) Trigonal bipyramidal
- (4) Square pyramidal

28) What is the oxidation number of Fe in $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ ion?

- (1) +2
- (2) +3
- (3) +1
- (4) 0

29) Geometrical isomerism in coordination compounds is exhibited by:

- (1) Square planar and tetrahedral complexes
- (2) Square planar and octahedral complexes
- (3) Tetrahedral and octahedral complexes
- (4) Square planar, tetrahedral and octahedral complexes

30) The colour of the coordination compounds depends on the crystal field splitting. What will be the correct order of absorption of wavelength of light in the visible region, for the complexes :



- (1) $[\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- (2) $[\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-}$
- (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-}$
- (4) $[\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$

31) The number of geometrical isomers and optical isomers of $[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{Py})]$ are respectively :

- (1) 2, 0
- (2) 3, 2
- (3) 3, 0
- (4) 2, 2

32) Which of the following does not show fac-mer isomerism?

- (1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (2) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3]$
- (3) $[\text{Co}(\text{en})_3]^{3+}$
- (4) All of these

33) CO_3^{2-} and SO_3^{2-} can be distinguished by :

- (1) $\text{K}_2\text{Cr}_2\text{O}_7$
- (2) KMnO_4
- (3) Lime water Test
- (4) Both (1) and (2)

34) Which of the following anions is not easily removed from aqueous solution by precipitation?

- (1) Cl^-
- (2) SO_4^{2-}
- (3) NO_3^-
- (4) CO_3^{2-}

35) Calculate the difference of crystal water present in Alum and Mohr salt :

- (1) 24
- (2) 10
- (3) 18
- (4) 16

1) Lead has been placed in group 1 and 2 both in qualitative analysis, because :

- (1) it shows the valency one and two
- (2) it forms insoluble PbCl_2
- (3) it forms lead sulphide
- (4) PbCl_2 formed in group 1 is partially soluble in water

2) Which does not dissolve in concentrated ammonium hydroxide solution ?

- (1) AgF
- (2) AgCl
- (3) AgBr
- (4) AgI

3) A white solid is first heated with dilute H_2SO_4 and then with concentrated H_2SO_4 . No action is observed in either case. The solid contains :

- (1) Sulphide
- (2) Sulphite
- (3) Sulphate
- (4) Chloride

4) The presence of magnesium is confirmed in the qualitative, analysis by the formation of a white crystalline precipitate of :

- (1) $\text{Mg}(\text{HCO}_3)_2$
- (2) $\text{Mg}(\text{NH}_4)\text{PO}_4$
- (3) $\text{MgNH}_4(\text{HCO}_3)_3$
- (4) MgCO_3

5) In Nessler's reagent for detection of ammonia, the active species is :

- (1) Hg_2Cl_2
- (2) Hg^{2+}
- (3) Hg_2I_2
- (4) HgI_4^{2-}

6)

	List-I (Basic Radical)		List-II (Group)
(P)	Pb^{+2}	(1)	VI group
(Q)	Fe^{+3}	(2)	IV group
(R)	Ni^{+2}	(3)	I group

(S)	Mg^{+2}	(4)	III group
-----	------------------	-----	-----------

Correct matching is :

	P	Q	R	S
(1)	1	3	2	4
(2)	3	4	2	1
(3)	4	3	2	1
(4)	3	4	1	2

(1) 1

(2) 2

(3) 3

(4) 4

7) Match the List-I with List-II

	Column-I (Cation)		Column-II (Group reagent)
A	$\text{Pb}^{+2}, \text{Cu}^{+2}$	i	$\text{H}_2\text{S} + \text{HCl}$
B	$\text{Al}^{+3}, \text{Fe}^{+3}$	ii	$(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{OH}$
C	$\text{Co}^{+2}, \text{Ni}^{+2}$	iii	$\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
D	$\text{Ba}^{+2}, \text{Ca}^{+2}$	iv	$\text{H}_2\text{S} + \text{NH}_4\text{OH}$

Correct match is.

(1) A-iv; B-ii; C-iii; D-i

(2) A-i; B-iii; C-iv; D-ii

(3) A-iii; B-i; C-iv; D-ii

(4) A-i; B-iii; C-ii; D-iv

8) Which is soluble in water.

(1) AgI

(2) BiI_3

(3) BaSO_4

(4) KNO_3

9) Which of the following cation form blue coloured soluble complex with excess of $\text{NH}_3(\text{aq.})$?

(1) Cr^{+3}

(2) Zn^{+2}

(3) Cu^{+2}

(4) Sn^{+2}

10) The number of d-electrons in Fe^{2+} is not equal to the number of electrons in which one of the following ?

- (1) d-electrons in Fe ($Z=26$)
- (2) p-electrons in Ne ($Z=10$)
- (3) s-electrons in Mg ($Z=12$)
- (4) p-electrons in Cl ($Z=17$)

11) Which of the following does not give chromyl chloride test ?

- (1) NaCl
- (2) KCl
- (3) NH_4Cl
- (4) HgCl_2

12)

Arrange Ce^{+3} , La^{+3} , Pm^{+3} and Yb^{+3} in increasing order of their ionic radii

- (1) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$
- (2) $\text{Ce}^{+3} < \text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3}$
- (3) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3}$
- (4) $\text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3} < \text{Yb}^{+3}$

13) Most common oxidation state of lanthanoid

- (1) +5
- (2) +3
- (3) +5
- (4) +7

14) Consider the oxidation number of the underlined atom in the following species. Identify the incorrect option:-

- (1) $\text{Cu}_2\underline{\text{O}}$ is -1
- (2) $\underline{\text{Cl}}\text{O}_3^-$ is +5
- (3) $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$ is + 6
- (4) $\text{H}\underline{\text{Au}}\text{Cl}_4$ is +3

15) Oxidation state of Mn in is +7 indicating all electrons paired in Mn but is coloured. It is due to

- (1) Charge transfer
- (2) Presence of unpaired electron in d-orbital in oxygen
- (3) Both (1) and (2)
- (4) None of the above

BOTANY

SECTION-1

1) XO-type of sex determination found in :-

- (1) *Drosophila*
- (2) Birds
- (3) Humans
- (4) Grasshopper

2) If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness ?

- (1) 100%
- (2) 25%
- (3) 50%
- (4) 75%

3) Chromosomal theory of inheritance was proposed by :-

- (1) Bateson & Punnet
- (2) Sutton & Boveri
- (3) T.H. Morgan
- (4) Watson & Crick

4) Law of dominance and law of segregation was proposed by Mendel, based on the observations of :-

- (1) Monohybrid cross
- (2) Dihybrid cross
- (3) Trihybrid cross
- (4) Back cross

5) In which of the following conditions the F_1 offsprings do **not** resemble either of the parents, but is inbetween the two parents ?

- (1) Complete dominance
- (2) Incomplete dominance
- (3) Co-dominance
- (4) Multiple allelism

6) Given below is distance between different genes on a chromosome

$$a - b = 21 \text{ cM}$$

$$b - c = 7 \text{ cM}$$

$$a - c = 14 \text{ cM}$$

$$a - d = 6 \text{ cM}$$

$$d - b = 15 \text{ cM}$$

Find the recombination frequency between gene c and d

- (1) 15 %
- (2) 8%
- (3) 20%
- (4) 21%

7) The percentage of 'ab' gametes produced by a parent with genotype AaBb will be:-

- (1) 12.5%
- (2) 25%
- (3) 50%
- (4) 75%

8) Which of the following characteristics represent 'Inheritance of blood groups' in humans?

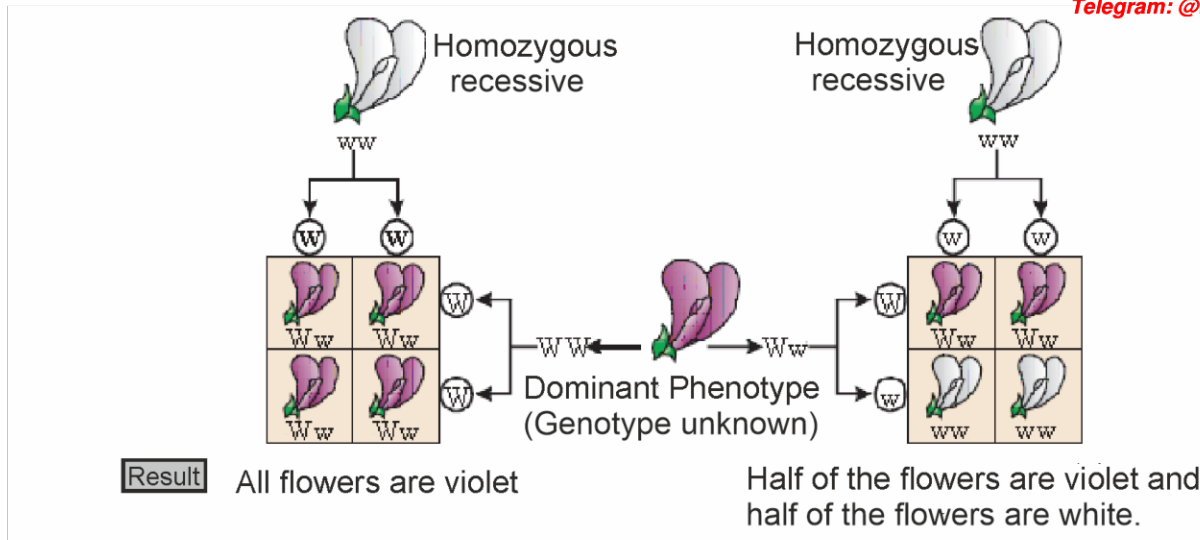
- a. Dominance
- b. Co-dominance
- c. Multiple allelism
- d. Incomplete dominance
- e. Polygenic inheritance

- (1) b, c and e
- (2) a, b and c
- (3) b, d and e
- (4) a, c and e

9) In a dihybrid cross in garden pea 1024 plants were obtained in F_2 . How many plants will show heterozygous trait for both characters ?

- (1) 218
- (2) 324
- (3) 512
- (4) 256

10) Observe the given diagram & select the **correct** option :-



- (1) It is the diagrammatic representation of a dihybrid cross.
- (2) It is the comparison of two monohybrid crosses, used to find out the phenotype of organisms.
- (3) It is the diagrammatic representation of reciprocal cross to find out the genotype of organism.
- (4) It is the diagrammatic representation of test cross.

11) The best example for pleiotropy is :-

- (1) Skin colour
- (2) Phenylketoneuria
- (3) Colour blindness
- (4) ABO blood group

12) In which of the following cases for a gene, the modified allele is said to be equivalent to the unmodified allele?

- (i) If it produces a non-functional enzyme
- (ii) If it produces less efficient enzyme
- (iii) If this allele does not produce any enzyme

- (1) (i) and (ii) only
- (2) (ii) only
- (3) (ii) and (iii) only
- (4) All (i), (ii) and (iii)

13) The ratio of total number of genotypes and phenotypes for ABO blood types in human being is

- (1) 3 : 2
- (2) 4 : 1
- (3) 3 : 1
- (4) 2 : 1

14) One of the example of polygenic inheritance is

- (1) Size of starch grains in pea seeds

- (2) Pattern baldness
- (3) Human skin colour
- (4) Synthesis of phenylalanine hydroxylase in human beings

15) How many of the following are sex linked disorders?

Thalassemia, Sickle-cell anaemia, Haemophilia, Turner's syndrome, Colour blindness, Myotonic dystrophy

- (1) Three
- (2) Four
- (3) Five
- (4) Two

16) Proportion of recombinant phenotypes in F_2 offsprings of mendelian dihybrid cross is:

- (1) $\frac{6}{16}$
- (2) $\frac{3}{16}$
- (3) $\frac{10}{16}$
- (4) $\frac{9}{16}$

17) How many types of gametes will be produced by a plant having genotype AaBbDdEE?

- (1) 16
- (2) 8
- (3) 4
- (4) 2

18) How many dominant traits among these were used by Mendel in his experiments ?

- (i) Round (ii) Dwarf
- (iii) Axial (iv) Inflated
- (v) White (vi) Constricted
- (vii) Terminal (viii) Violet
- (ix) Tall

- (1) Five
- (2) Eight
- (3) Three
- (4) Nine

19) 20 types of amino acids are coded by how many total sense codons?

- (1) 10
- (2) 61

(3) 30

(4) 64

20) In bacteria AAA codon codes for lysine amino acid then in human AAA codon codes for :-

(1) Glycine

(2) Phenyl alanine

(3) Leusine

(4) Lysine

21) In secondary structure of t-RNA it look like a :-

(1) Inverted L

(2) Clover leaf

(3) Globular

(4) Fibrous

22) The sequence of which chromosome number was completed in May 2006

(1) Chromosome number 1

(2) Chromosome number 2

(3) Chromosome number 5

(4) Chromosome number 10

23) Severo ochoa enzyme is :-

(1) DNA polymerase

(2) Polynucleotide phosphorylase

(3) DNA ligase

(4) Helicase

24) Cistron is a :-

(1) A segment of DNA coding for a polypeptide

(2) A segment of DNA coding for a t RNA

(3) A Segment of DNA coding for r-RNA

(4) A segment of DNA codins for s-RNA

25) The average rate of polymerisation in E.coli is approximately :-

(1) 2000 base pairs per second

(2) 2000 bases per minute

(3) 2000 bases per hour

(4) 20000 base pairs per second

26) The flow of genetic information from DNA to mRNA and then to protein is called :-

- (1) Transcription
- (2) Translation
- (3) Central dogma
- (4) Genetic code

27) Which of the following RNA is known as adapter RNA ?

- (1) mRNA
- (2) tRNA
- (3) rRNA
- (4) Sn RNA

28) In the following representation A and B is - *Streptococcus pneumoniae* S Strain → injected into Mice → A

Streptococcus pneumoniae R Strain → injected into Mice → B

- (1) A → Mice die
B → Mice live
- (2) A → Mice die
B → Mice die
- (3) A → Mice live
B → Mice live
- (4) A → Mice live
B → Mice die

29)

RNA polymerase I transcribe

- (1) 5S rRNA
- (2) 18S rRNA
- (3) 23S rRNA
- (4) tRNA

30) A molecule that can act as genetic material must **not** have following properties :-

- (1) It should be chemically and structurally highly reactive.
- (2) It should be able to generate its replica.
- (3) It should provide the scope for slow changes.
- (4) It should be able to express itself in the form of Mendelian characters

31) Taylor demonstrated semiconservative chromosomal replication in :-

- (1) *Vicia faba*
- (2) *Pisum sativum*
- (3) Liver cell
- (4) Muscle cell

32) *Lac* operon is turned on when allolactose binds to :-

- (1) Operator gene
- (2) mRNA
- (3) Repressor protein
- (4) Promoter site

33) Select the **correct** sequences of steps in DNA finger printing :

- (I) Hybridisation using labelled VNTR probe.
- (II) Isolation of DNA
- (III) Transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon
- (IV) Detection of hybridised DNA fragments by autoradiography
- (V) Separation of DNA fragments by electrophoresis
- (VI) Digestion of DNA by restriction endonucleases.

- (1) I, V, VI, II, III and IV
- (2) II, VI, V, III, I and IV
- (3) V, I, VI, III, IV and II
- (4) II, I, V, VI, IV, III

34) The variable number of tandem repeats (VNTR) belongs to a class of satellite DNA referred to as:

- (1) SNP
- (2) Mini-satellite
- (3) Microsatellite
- (4) Sequence annotation

35) Human Genome Project was closely associated with the rapid development of a new area in biology called :-

- (1) Genomics
- (2) Genetic Engineering
- (3) Bioinformatics
- (4) Biotechnology

SECTION-2

1) **Assertion (A)** : A father will be never pass the gene for haemophilia to his sons.

Reason (R) : Haemophilia is x-linked recessive traits.

- (1) (A) is correct (R) is not correct
- (2) (A) is not correct but (R) is correct
- (3) Both (A) & (R) correct and (R) is the correct explanation of (A)
- (4) Both (A) & (R) are correct but (R) is not the correct explanation of (A)

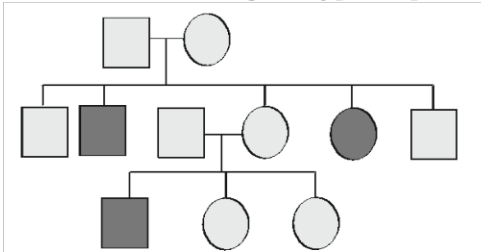
2) Chromosomal map prepared by :-

- (1) Sutton & Boveri
- (2) Bateson & Punnett
- (3) Morgan
- (4) Sturtevant

3) Which of the following couple can claim that the baby with 'O' blood group type are possibly the biological parents of it?

- (1) AB and A
- (2) A and O
- (3) O and AB
- (4) B and AB

4) Given pedigree chart shows inheritance of autosomal recessive trait (for eg - sickle cell anaemia) then what will be genotype of parents?



- (1) Father - (Aa) Mother - (aa)
- (2) Father - (aa) Mother - (aa)
- (3) Father - (Aa) Mother - (Aa)
- (4) Father - (AA) Mother - (AA)

5) In the cross $AaBb \times Aabb$, the genotype $AABb$: $AaBB$: $aabb$ and $aaBb$ will be obtained in the ratio :-

- (1) 1 : 0 : 2 : 1
- (2) 2 : 1 : 4 : 0
- (3) 1 : 0 : 1 : 1
- (4) 1 : 2 : 1 : 2

6) A comparison between **A** and **B** is given. Identify **A** and **B** :

A	B
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete.	Segregate at gamete formation and only one of each pair is transmitted to a gamete.
Independent pairs segregate independently of each other.	One pair segregates independently of another pair.

Options	A	B
(1)	Chromosome	Linked genes
(2)	Linked genes	Chromosome
(3)	Gene	Chromosome
(4)	Chromosome	Gene

- (1) 1
(2) 2
(3) 3
(4) 4

7) In *Antirrhinum species* the F_1 progeny (pink flower) is self crossed, than find the **correct** statement among the following:-

- (1) 50% plants will be tall
(2) 50% plants will have Red flower
(3) 25% plants will have white flower
(4) 25% plants will have pink flower

8) Match the column **correctly** :-

	Column-I		Column-II
(i)	Hershey & Chase	(a)	Deciphering of genetic code
(ii)	Avery, Macleod and McCarty	(b)	Genetic material in bacteriophage is DNA
(iii)	Marshall Nirenberg	(c)	Concept of adapter molecule
(iv)	Francis Crick	(d)	In transforming principle of Griffith substance was DNA

- (1) (i)-(a), (ii)-(d), (iii)-(b), (iv)-(c)
(2) (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)
(3) (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)
(4) (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a)

9) DNA dependent DNA polymerase enzyme uses

- (1) Deoxyribonucleoside triphosphate as a substrate
(2) Ribonucleoside triphosphate as a substrate
(3) Both (1) & (2)

(4) RNA template for catalyzing polymerization of nucleotide of DNA

10) Refer the given mRNA segment. It can be translated completely into a polypeptide. Which of the

following codons may correspond with **A** and **B** 5'

A		

 CGU GUU AUC

B		

 3'

- (1) (A) AUG, (B) UAA
- (2) (A) UAA, (B) AUG
- (3) (A) AUG, (B) GUG
- (4) (A) AUG, (B) UAC

11)

Select the **incorrect** statements

- (1) A typical nucleosome contains 200 bp of DNA helix
- (2) Histones are negatively charged basic proteins
- (3) Euchromatins are transcriptionally active
- (4) Heterochromatin is more densely packed

12) If a hybrid DNA " $N^{15} N^{14}$ " is allowed to replicate in N^{14} medium, after three generations how many DNA will be having " $N^{15} N^{14}$ " :

- (1) 3
- (2) 4
- (3) 8
- (4) 1

13) Which of following statement is **true** for *lac* operon in *E-coli*.

- (a) *lac* operon consists of one regulatory gene and five structural gene.
- (b) The **i** gene codes for inducer
- (c) The **z**-gene codes for β -galactosidase
- (d) A very low level of expression of *lac* operon has to be present in cell all the time

- (1) a, b, c, d
- (2) a, c, d
- (3) c, d
- (4) b, d

Victim	Crime scene sample	Suspect-1	Suspect-2
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

14) Which suspect would you charge with the crime?

- (1) Both suspect 1 & 2

- (2) Only suspect 1
- (3) Only suspect 2
- (4) Neither suspect 1 nor suspect 2

15) Find an **incorrect** statement with respect to finding of HGP :-

- (1) Dystrophin gene is largest gene of human genome.
- (2) Chromosome-Y has fewest genes (231).
- (3) SNP's were identified at about 2.4 million location.
- (4) Less than 2 percent of genome codes for proteins

ZOOLOGY

SECTION-1

1) Which of the following is a blood cholesterol lowering agent, produced by yeast *Monascus purpureus* ?

- (1) Cyclosporin-A
- (2) Statins
- (3) Streptokinase
- (4) Amylase

2) The puffed-up appearance of dough is due to :-

- (1) Growth of LAB
- (2) Production of O₂ & ethanol
- (3) Production of CO₂ gas
- (4) Growth of *Monascus* yeast

3) Read the following statements :-

- (A) Cyclosporin A is used as a clot buster
- (B) Whisky, brandy & rum are produced by distillation of fermented broth
- (C) The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*
- (D) The biogas production technology in India is developed by ICAR

Which of the above statements are not **incorrect** ?

- (1) A & B
- (2) B & C
- (3) C & D
- (4) A & D

4) Which one of the following antibiotic was extensively used to treat American soldiers wounded in World War-II:-

- (1) Streptokinase
- (2) Penicillin
- (3) Statins
- (4) Neomycin

5) Full potential of penicillin as an effective antibiotic was established by :-

- (1) Ernest Chain
- (2) Howard Florey
- (3) Calvin
- (4) Both 1 and 2

6) *Propionibacterium sharmanii* is used in the ripening of which cheese ?

- (1) Camembert cheese
- (2) Swiss cheese
- (3) Roquefort cheese
- (4) Limburger cheese

7) **Statement I:** Cyanobacteria increases the soil fertility and also add the organic matter to the soil.
Statement II: They can fix atmospheric nitrogen.

- (1) Both statements I and II are correct
- (2) Both statements I and II are incorrect
- (3) Only statement I is correct
- (4) Only statement II is correct

8) Which of the following is **not** an autotrophic bacteria which fix atmospheric nitrogen?

- (1) Anabaena
- (2) *Azotobacter*
- (3) *Nostoc*
- (4) *Oscillatoria*

9) Statement A: Lactic Acid Bacteria (LAB) are responsible for converting milk into curd.
Statement B: LAB creates acidic medium necessary to coagulate and fully digest the milk proteins.

- (1) Statements A and B are correct
- (2) Statements A and B are incorrect.
- (3) Statement A is correct but B is incorrect
- (4) Statement B is correct and A is incorrect.

10) Cyclosporin A produced by the fungus

- (1) *Saccharomyces cerevisiae*
- (2) *Albugo*

(3) *Trichoderma polysporum*

(4) *Monascus purpureus*

11) The vitamin whose content increases following the conversion of milk into curd by lactic acid bacteria is :

(1) vitamin C

(2) vitamin D

(3) vitamin B₁₂

(4) vitamin E

12) What is B.O.D.

(1) The amount of O₂ utilised by organism in water

(2) The amount of O₂ that would be consumed if all the organic matter in one litre of water were oxidised by bacteria.

(3) The total amount of O₂ present in water

(4) All of the above

13)

Match the items of column-I with column-II and choose correct answer.

	Column-I		Column-II
(A)	Lady bird	(I)	<i>Methanobacterium</i>
(B)	Mycorrhiza	(II)	<i>Trichoderma</i>
(C)	Biological control	(III)	Aphids
(D)	Biogas	(IV)	<i>Glomus</i>

(1) A-III; B-II; C-I; D-IV

(2) A-IV; B-I; C-II; D-III

(3) A-III; B-IV; C-II; D-I

(4) A-II; B-IV; C-III; D-I

14) The technology of biogas production was developed in India mainly due to the efforts of :-

(1) Indian Council of Agricultural Research (ICAR).

(2) Indian Council of Medical Research (ICMR).

(3) National Botanical Research Institute (NBRI).

(4) Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC).

15) Which of the following disease caused internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage ?

(1) Ascariasis

- (2) Filariasis
- (3) Amoebiasis
- (4) Trypanosomiasis

16) Physiological barrier which prevents entry of microorganisms in our body is/are

- (1) Acid in the stomach
- (2) saliva in the mouth
- (3) tears from eyes
- (4) All of the above

17) Antigen binding site in an antibody is found between

- (1) Two light chains
- (2) Two heavy chains
- (3) One heavy and one light chain
- (4) Either between two light chains or between one heavy and one light chain depending upon the nature of antigen

18) Allergens are

- (1) Infectious and increased secretion of IgE
- (2) Non-infectious and increased secretion of IgE
- (3) Infectious and increases secretion of IgG
- (4) Non-infectious and increased secretion of IgM

19) How does vaccination work ?

- (1) The immune system produces antibodies, which stay in the blood
- (2) Memory lymphocytes are produced. They remain in the body to fight off any future infection with the live pathogen
- (3) The dead pathogen stays in the body and constantly stimulates the immune system
- (4) All of the above

20) Which form of tumour remain confined to their original location and do not spread to other parts the body ?

- (1) Malignant tumour
- (2) Benign tumour
- (3) Both (a) and (b)
- (4) Leukaemia

21) Both ___A___ and ___B___ provide micro environments for development and maturation of ___C___. Choose A, B, C from given options.

	A	B	C
(1)	Bone marrow	Thymus	Leucocyte
(2)	Spleen	Thymus	Lymphocytes
(3)	Bone marrow	Thymus	Lymphocytes
(4)	Lymphnodes	Bone marrow	Lymphocytes

- (1) 1
- (2) 2
- (3) 3
- (4) 4

22) Which of the following two statements correct for kidney transplant ?

- (a) Immuno suppressant drugs used for long time inspite kidney properly transplanted
- (b) Graft rejection is due to cell mediated immunity
- (c) B-cells responsible for graft rejection
- (d) Acceptance and rejection of graft depends on specific interferons.

- (1) (a) and (b)
- (2) (b) and (c)
- (3) (c) and (d)
- (4) (a) and (c)

23) **Assertion** : Innate immunity is a non-specific type of deffence.

Reason : This is accomplished by providing different types of barriers to the entry of foreign agents into our body.

- (1) Assertion & Reason correct, Reason is correct explanation to Assertion
- (2) Assertion & Reason correct, Reason is not correct explanation to Assertion
- (3) Assertion is correct, Reason is incorrect
- (4) Assertion & Reason are incorrect

24) Cellular barriers of innate immunity included all the following except:

- (1) N K cells
- (2) Monocytes
- (3) PMNL
- (4) B lymphocytes

25)

Tobacco consumption is known to stimulates secretion of adrenaline and nor-adrenaline. The component causing this could be:

- (1) Nicotine
- (2) Tannic acid
- (3) Curamin

(4) Catechin

26) Ratio of Methane, Ammonia, Hydrogen in Stanley Miller's experiment was :-

- (1) 3 : 1 : 2
- (2) 2 : 1 : 2
- (3) 1 : 2 : 1
- (4) 5 : 4 : 1

27) Variations in gene frequencies occur by chance in:-

- Small population · Founder effect
- Bottleneck effect · Natural selection

How many of the above statements are applicable to genetic drift ?

- (1) 3
- (2) 4
- (3) 2
- (4) 1

28) Industrial melanism, drug resistance are examples of :-

- (1) Stabilising selection
- (2) Directional selection
- (3) Disruptive selection
- (4) Artificial selection

29) The early atmosphere of earth was reducing as it was composed mostly of ?

- (1) Hydrogen
- (2) Helium
- (3) Oxygen
- (4) Methane

30) Presence of which of the following factors will not cause deviation in the gene frequency of a large population as defined by Hardy-Weinberg equilibrium ?

- (A) Recombination
- (B) Random mating
- (C) Mutation
- (D) Genetic drift

Options are :-

- (1) Only (A) & (B)
- (2) Only (B)
- (3) Only (B) & (D)
- (4) Only (D)

31) A = Natural selection

B = Variation & their inheritance

C = Survival of fittest

D = Struggle for existence

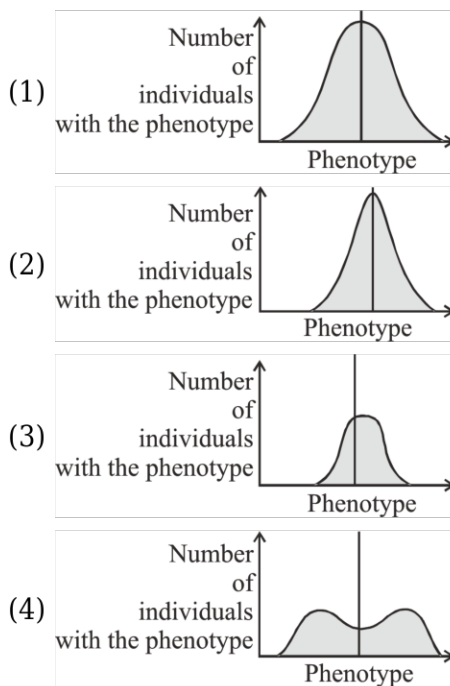
according to darwinism, correct sequence are:-

- (1) B, C, A, D
- (2) D, B, C, A
- (3) C, D, A, B
- (4) D, A, C, B

32) The first life on earth originated from nonliving materials has been explained by :-

- (1) Theory of biogenesis
- (2) Theory of abiogenesis
- (3) Theory of special creation
- (4) Theory of extra-terrestrial origin

33) The diagrammatic representation of operation of natural selection is shown. Which of the following option indicates 'directional selection' ?



34) In a large population frequency of recessive allele is 0.7, then frequency of homozygous dominant organisms is:-

- (1) 0.09
- (2) 0.49
- (3) 0.3
- (4) 0.7

35) Dryopithecus and Ramapithecus were existing before how many years :-

- (1) 50 million years
- (2) 60 million years
- (3) 15 million years
- (4) 45 million years

SECTION-2

1) According to Hugo de vries, single step large mutation is called

- (1) sport
- (2) saltation
- (3) variation
- (4) mutation

2) In galapagos island, Darwin found different varieties of finches, originated from a common ancestor. The original variety of these finches was:-

- (1) Insectivorous
- (2) Vegetarian
- (3) Seed-eating
- (4) Cactus eating

3) Sweet potato and potato are modification of respectively :-

- (1) Root and Stem
- (2) Stem and Root
- (3) Root and Leaf
- (4) Root and Root

4) Common ancestor of Tuataras and Crocodiles is :-

- (1) Dinosaurs
- (2) Thecodonts
- (3) Synapsids
- (4) Sauropsids

5) According to theory of spontaneous generation?

- (1) Life came from pre-existing life
- (2) Life came from both living and non-living matter
- (3) Life came from pre-existing, non-living, organic molecules
- (4) Life originated from decaying and rotting matter like straw, mud etc.

6) Read the following statements :

- (i) Transmission of HIV infection can occur from infected mother to her child through placenta.
- (ii) Drug addicts who take drugs by snorting are at high risk of getting HIV infection

(iii) Macrophage acts like an HIV factory

(iv) Treatment of AIDS with antiretroviral drugs is completely effective.

How many of the given statement/s is/are correct:

- (1) One
- (2) Two
- (3) Three
- (4) Four

7) Virus infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. They are a part of :-

- (1) Physical barriers
- (2) Physiological barriers
- (3) Cellular barriers
- (4) Cytokine barriers

8) Fill in the different columns of the table given below :-

	Disease	Causal organism and part it inputs	Medium of transfer
i	Common cold	A	Droplets from sneezing of infected persons
ii	B	<i>Trichophyton</i> skin, nails and scalp	Using towels of infected individuals
iii	Pneumonia	C	Droplets/aerosols released by an infected person sharing utensils
iv	Filariasis	<i>Wucheria</i> , Lymphatic vessels	D

In which one of the following

options all the blanks A, B, C and D are Correctly filled.

- (1) A-Rhinovirus. Nose and respiratory passage not lungs
B-Ringworm, C - Haemophilus influenzae. Alveoli of lungs, D - Contaminated food and water
- (2) A-Coryza virus. Alveoli of lungs, B-Ringworm, C-Streptococcus pneumoniae. Nose and respiratory passage, D - Culex mosquito
- (3) A-Rhinovirus. Nose and respiratory passage not lungs, B-Ascariasis, C-Haemophilus influenzae Lungs, D-Culex mosquito
- (4) A-Rhinovirus. Nose and respiratory passage not lungs, B-Ringworm, C-Haemophilus influenzae. Alveoli of lungs, D-Culex mosquito

9) Statement - I : Houseflies act as mechanical carriers in amoebic dysentery.

Statement - II : Amoebiasis, typhoid and ascariasis are transmitted by contaminated food and water.

- (1) Both statement-I and statement-II are correct.
- (2) Both statement-I and statement-II are incorrect.
- (3) Statement-I is incorrect and statement-II is correct.
- (4) Statement-I is correct and statement-II is incorrect.

10) The medicines used to help patients cope with mental illness and insomnia

- (1) Barbiturates
- (2) Amphetamines
- (3) Benzodiazepines
- (4) All of these

11) In an allergic reaction, which chemicals are released by the mast cells?

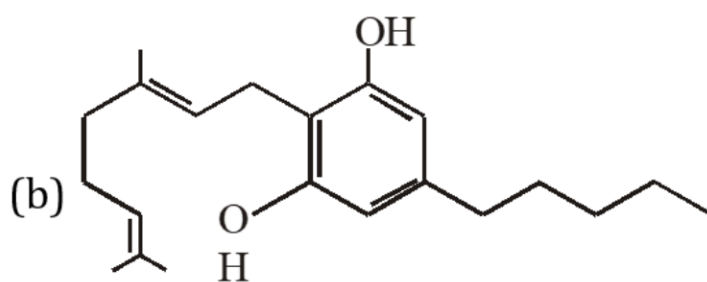
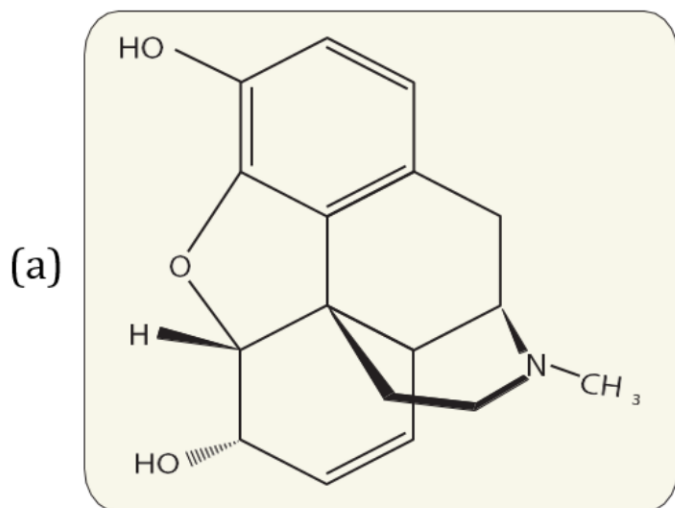
- (1) Histamine and serotonin
- (2) Histamine and secretin
- (3) Serotonin and adrenalin
- (4) Serotonin and nor-adrenalin

12) Neoplastic transformation by DNA damage can be brought about by several factors. Choose the factors which lead to oncogenic transformation within exposed cells :-

- (a) U.V. rays
- (b) X-rays
- (c) Gamma-rays
- (d) Tobacco smoke
- (e) Retrovirus

- (1) a, b, c
- (2) a, b, c, d
- (3) a, b, d
- (4) b, d

13) Identify the molecules (a) and (b) given below and select the right option giving their source and use.



	Molecule	Source	Use
(1)	(a) Cocaine	<i>Erythroxylum coca</i>	Accelerates the transport of dopamine
(2)	(b) Heroin	<i>Cannabis sativa</i>	Depressant and slows down body functions
(3)	(b) Cannabinoid	<i>Atropa belladonna</i>	Produces hallucinations
(4)	(a) Morphine	<i>Papaver somniferum</i>	Sedative and pain killer

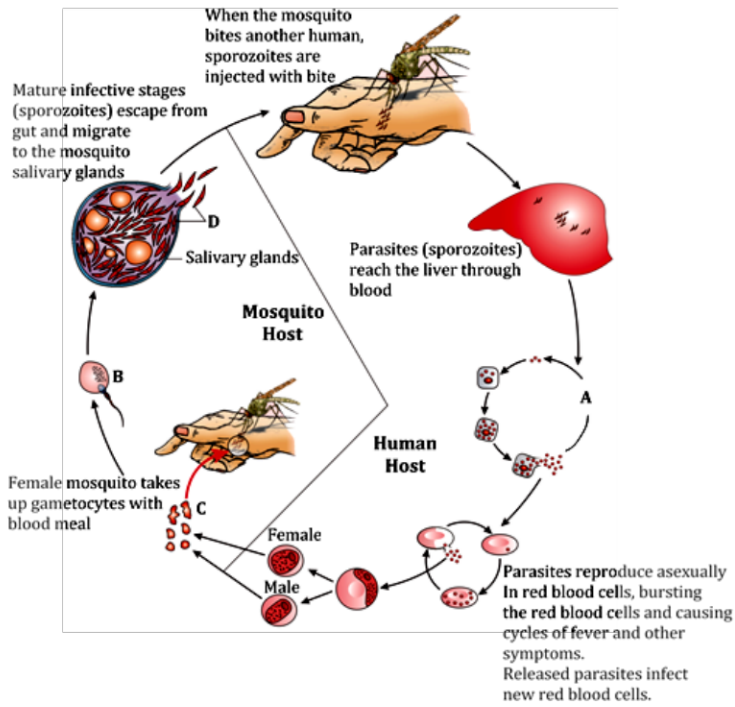
(1) 1

(2) 2

(3) 3

(4) 4

14) Refer to the given figure showing stages in the life cycle of use plasmodium. In the figure, which type of reproduction is occurring at stages A and B respectively. Also, identify C and D in the figure.



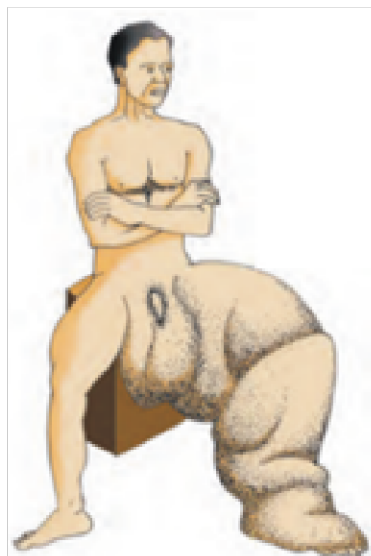
	A	B	C	D
(1)	Asexual	Sexual	Gametocytes	Sporozoites
(2)	Sexual	Asexual	Gametocytes	Sporozoites
(3)	Asexual	Sexual	Sporozoites	Gametocytes
(4)	Asexual	Sexual	Gametocytes	Sporozoites

(1) 1

(2) 2

(3) 3

(4) 4



15) Refer to the given figure.

Which disease is shown in the figure?

- (1) Amoebiasis
- (2) Filariasis
- (3) Elephantiasis

(4) Both (2) and (3)

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	1	2	1	3	1	2	3	4	2	3	1	4	4	4	1	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	4	4	1	1	2	3	1	4	2	2	3	4	2	3	4					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	1	2	1	2	2	2	3	1	2	1	1	4	1	2	4

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	4	2	4	3	2	2	4	2	2	2	2	3	1	4	4	3	3	3	3	4
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	1	3	4	1	3	3	3	3	2	3	3	3	4	3	3					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	4	4	3	2	4	2	2	4	3	4	4	1	2	1	1

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	1	2	1	2	2	2	2	4	4	2	2	1	3	4	1	2	1	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	2	1	2	1	1	3	2	1	2	1	1	3	2	2	3					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	2	3	3	3	3	2	1	1	2	4	3	3	3

ZOOLOGY

SECTION-1

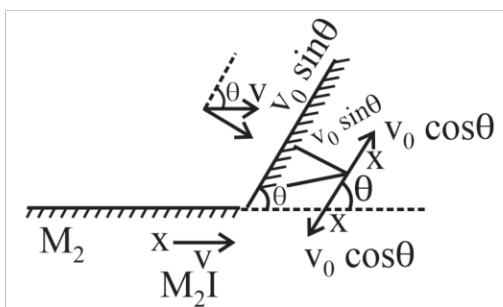
Telegram: @NEETxNOAH

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	2	2	4	2	1	2	3	3	3	2	3	4	1	4	3	2	2	2
Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185					
A.	3	1	2	4	1	2	1	2	1	2	2	2	3	1	3					

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	2	3	1	4	4	2	4	4	1	4	1	2	4	1	4

PHYSICS



3)

Relative to M_2

$$V_{\text{net relative } M_2} = 2 v_0 \sin \theta$$

$$5) \sin i_c = \frac{1}{\mu} \propto \lambda$$

$$i_c \propto \lambda$$

Yellow, orange, red emerge from air.

$$6) d = \frac{t}{\cos r} \sin(i - r)$$

For small angles $\sin(i - r) \approx i - r$; $\cos r \approx 1$

$$d = t(i - r), d = it \left[1 - \frac{r}{i} \right]$$

9)

second image from L2 is at 5m from mirror

distance from person = 5+1 = 6m

$$10) \text{ Radius of circle, } r = \frac{h}{\sqrt{\mu^2 - 1}} = \frac{12}{\sqrt{\left(\frac{4}{3}\right)^2 - 1}} = \frac{12 \times 3}{\sqrt{7}} \text{ cm}$$

11)

$$\mu = \frac{\sin\left(\frac{30^\circ + 60^\circ}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)} = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2}$$

$$12) m = m_0 \times m_e m$$

$$= m_0 \times \left(1 + \frac{D}{f_e}\right)$$

$$\Rightarrow 100 = 10 \times \left(1 + \frac{25}{f_e}\right) \Rightarrow f_e = \frac{25}{9} \text{ cm}$$

13)

$$f_0 = 7 \text{ mm}, f_e = 20 \text{ mm} \quad v_0 + u_e = 20 \text{ cm}$$

$$\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e} \quad v^0 + \frac{500}{27} = 200 \text{ mm}$$

$$\frac{1}{20} = \frac{1}{-250} - \frac{1}{u_e} \quad v^0 = \frac{4900}{27} \text{ mm}$$

$$u_e = \frac{-500}{27} \text{ mm}$$

$$m = m^0 \times m_e = \left(\frac{f_0 - v_0}{f_0}\right) \times \left(\frac{f_e}{f_e + u_e}\right)$$

$$\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \text{ or } \cot \frac{A}{2} = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

14) As

$$\frac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \text{ or}$$

$$\sin\left(\frac{\pi}{2} - \frac{A}{2}\right) = \sin\left(\frac{A}{2} + \frac{\delta}{2}\right);$$

$$\frac{\pi}{2} - \frac{A}{2} = \frac{A}{2} + \frac{\delta}{2}$$

$$\therefore \delta = \pi - 2A = 180^\circ - 2A$$

15)

Conceptual

17) 11 : 60

-04 : 50

7 : 10 AM

$$18) \text{ from one side } d_{\text{app}} = \left(\frac{1}{1.5}\right) x_1$$

$$x_1 = 5 \times 1.5 = 7.5 \text{ cm}$$

$$\text{from other side } d_{\text{app}} = \left(\frac{1}{1.5}\right) x_2$$

$$x_2 = 1.5 \times 3 = 4.5 \text{ cm}$$

$$\text{thickness} = 4.5 + 7.5 = 12 \text{ cm}$$

19) Since the ray emerges normally, therefore $e = 0$. According to relation $A + \delta = i + e$, we get $i = A + \delta$.

Hence by $\delta = (\mu - 1)A$, we get $i = \mu A$.

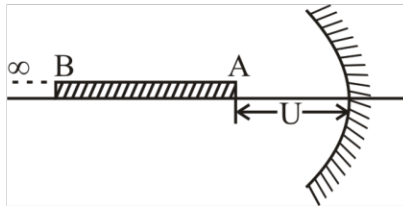
20)

For refraction through curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

$$\Rightarrow \frac{1.5}{v} - \frac{1}{(-15)} = \frac{(1.5 - 1)}{30}$$

$$\Rightarrow v = -30 \text{ cm}$$



22)

$$U_A = -U$$

$$f = -f$$

$$v = \frac{uf}{u - f} = \frac{-ux - f}{-u + f}$$

$$v = \frac{uf}{f - u}$$

$$U_B = -\infty$$

$$V_B = -f$$

$$\text{Length of image} = \frac{uf}{u - f} - f$$

$$= \frac{f^2}{u - f}$$

23) By refraction through curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

$$\Rightarrow \frac{1}{v} - \frac{4/3}{-6} = \frac{(1 - 4/3)}{-10}$$

$$\Rightarrow \frac{1}{v} = \frac{30}{17} - \frac{2}{9} \Rightarrow \frac{1}{v} = \frac{3 - 20}{90}$$

$$\Rightarrow \frac{1}{v} = -\frac{17}{90}$$

$$\Rightarrow v = 5.2 \text{ cm}$$

27)

$$f = \frac{R}{2(\mu - 1)} = \frac{40}{2(1.5 - 1)} = 40 \text{ cm}$$

28) Height of object = 25 cm,

height of image = 50 cm,

focal length of convex lens is 30 cm.

Let distance of real object from lens is x , and distance of real image from lens is y .

$$\frac{y}{x} = \frac{h_i}{h_o} = \frac{50}{25} = 2 \Rightarrow y = 2x$$

here $v = 2x$, $u = -x$, $f = 30$ cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{2x} - \frac{1}{-x} = \frac{1}{30}$$

$$\Rightarrow \frac{1}{2x} = \frac{1}{30} \Rightarrow x = 15 \text{ cm}$$

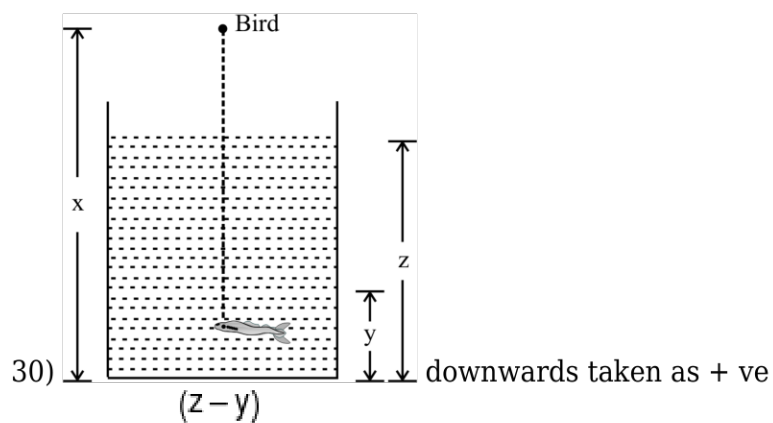
distance between real object and real image is

$$x + y = 15 + 2 \times 15 = 45 \text{ cm.}$$

29)

$$\text{Velocity of image} = - \left(\frac{f}{f - u} \right)^2 \text{ vel. of object}$$

$$\begin{aligned} f = 20 \text{ cm} = 0.2 \text{ m} &= - \left(\frac{+0.2}{+0.2 + 2.8} \right)^2 \times 15 \\ &= - \left(\frac{0.2}{3} \right)^2 \times 15 \\ &= - \left(\frac{1}{15} \right)^2 \times 15 = - \frac{1}{15} \text{ m/s} \end{aligned}$$



$$I = (x - z) + \frac{\mu}{3}$$

$$\frac{dI}{dt} = \frac{dx}{dt} - \frac{dz}{dt} + \frac{1}{\mu} \left(\frac{dz}{dt} - \frac{dy}{dt} \right)$$

$$V_I = (6 - 2) + \frac{3}{4} [2 - (-8)]$$

$$V_I = 4 + \frac{3}{2} \times 10 = \frac{23}{2} \text{ m/s}$$

31)

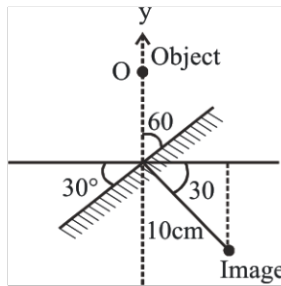
$$m = \frac{f_o}{f_e} = \frac{\beta}{\alpha}$$

$$\frac{100}{2} = \frac{\beta}{0.5} \Rightarrow \beta = 25^\circ$$

$$32) \quad \text{D} + \text{J} = \text{D}$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f_{eq.}}$$

$$(\mu_1 - 1)\bar{R} + (\mu_2 - 1)\bar{R} = \frac{1}{f_{eq.}} \Rightarrow f_{eq.} = \frac{R}{\mu_1 - \mu_2}$$



34) Co-ordinate of image $(10 \cos 30, -10 \sin 30)$
 $(5\sqrt{3}, -5)$

$$35) \quad \frac{f_1}{f_a} = \frac{(a\mu_g - 1)}{(l\mu_g - 1)} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{3/2}{5/4} - 1\right)} = \frac{5}{2}$$

$$f_1 = f_a \left(\frac{5}{2}\right) = \frac{12 \times 5}{2} = 30 \text{ cm}$$

37) Newton's formulae $f^2 = x_1 x_2$ or $f = \sqrt{x_1 x_2}$

45) $m = +ve \Rightarrow$ virtual image

$m = -ve \Rightarrow$ real image

$|m| > 1 \Rightarrow$ magnified image

$|m| < 1 \Rightarrow$ diminished image

47)

Shift by mirror

$$= t \left(1 - \frac{1}{\mu}\right)$$

$$= 5 \left(1 - \frac{1}{3}\right) = \frac{5}{3} \text{ cm}$$

Apart distance
from the mirror

$$10 + 5 - \frac{5}{3} = \frac{40}{3} \text{ cm}$$

Opposite side

apparent distance

$$\frac{40}{3} - \frac{5}{3} = \frac{35}{3} \text{ cm}$$

distance of final image
from the object is

$$15 + \frac{35}{3} = \frac{80}{3} \text{ cm}$$

49)

$$\mu_1 \sin i = \mu_2 \sin i \dots\dots\dots(1)$$

$$\mu_2 \sin r^1 = \mu_3 \sin r \dots\dots\dots(2)$$

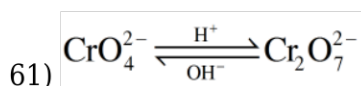
from (1) + (2)

$$\mu_1 \sin i = \mu_3 \sin r$$

$$\frac{\sin i}{\sin r} = \frac{\mu_3}{\mu_1}$$

Ans. 2

CHEMISTRY



95)

Fe²⁺: d-electrons : 6

Fe: p-electrons : 6

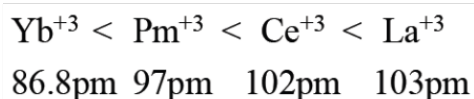
Ne: s-electrons : 6

Mg: p-electrons : 6

Cl: p-electrons : 11

97)

In lanthanides there is a regular decrease in the atomic radii as well as ionic radii of trivalent ions as the atomic number increases from Ce to Lu. This decrease in size of atoms and ions is known as **lanthanide contraction**. Although the atomic radii do show some irregularities but ionic radii decreases from La to Lu. Thus the correct order is.



99)

Ox. state of "O" in Cu₂O = -2

Ox. state of "Cl" in ClO₃⁻ = +5

Ox. state of "Cr" in $K_2Cr_2O_7 = +6$

Ox. state of "Au" in $HAuCl_4 = +3$

□ (1) is incorrect

100)

Fact based

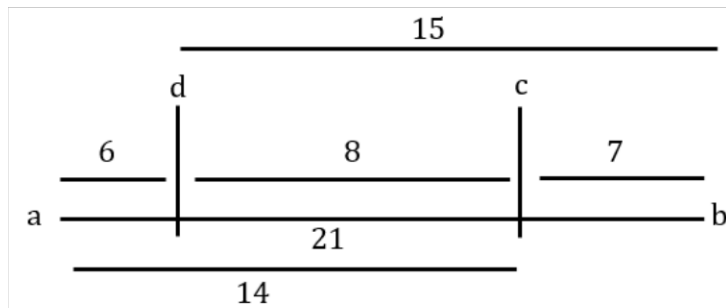
BOTANY

104) Results of monohybrid cross is the base of law of dominance and law of segregation.

NCERT Page No. 75

105) Incomplete dominance is the condition in which offsprings have intermediate phenotype

NCERT Page No. 76



106)

→ d to b = 15 cM

→ c to b = 7 cM

So, → D and C = 15 - 7 = 8 cM

1 cM, = 1%

So, 8 cM = 8%

107) Based on Dihybrid cross given in NCERT Page - 80

108) Allen Module Page No. 73

111)

NCERT XII Pg. No.#85

126)

NCERT-XII Pg. # 82

128) NCERT-XII, Pg. # 84

129)

NCERT XI Pg. # 151 Fig. 9.5

130) NCERT (XIth) Pg. # 103, Para # 2

139) In autosomal recessive disorders both unaffected parents can have affected child so, both parents are heterozygous.

143) NCERT-XII, Pg. # 98

147)

NCERT XII, Pg. # 89

150) NCERT XII, Page # 118-119

ZOOLOGY

152) NCERT-XII, Pg. # 151

153) NCERT XII Pg. # 152, 153, 156

154) NCERT (XII) Pg. # 182

155) NCERT XII, Pg. # 182

156) NCERT (XII) Pg. # 181

158) NCERT-XII, Page # 158

161) Collagen is the most abundant protein in animals whereas RuBisCO is the most abundant protein in whole of the biosphere. Lectins are secondary metabolites present in plants.

163)

NCERT XII Page # 157,155

164)

NCERT XII Page # 156

173) NCERT Page No. 134

174) Ncert Pg No: 134

175) Ncert Exemplar

176) NCERT (XIth) Pg. # 127

183) NCERT XII, Pg.# 137

190) NCERT XIIth Pg.#127